

SIT225 Data Capture Technologies

Credit Task: Live smooth Plotly Dash update for smartphone accelerometer data

Estimated time: 5-10 hours

Last updated: 14 June 2026

Due: End of Week 8, no later than week 9

Submit early, by week 8, to receive feedback early.

Instructions and Conditions

This task is due in **Week 9 (Sunday)**, and you are expected to start early. If your first submission is incomplete or incorrect, you may resubmit after receiving feedback. There will be **no extensions**, and late submissions will not be accepted. This task is a Credit task, failure to submit a correct version on time will result in not availing a Credit grade in the unit.

All submissions will be checked for **plagiarism** and **GenAI** use. *Any GenAI usage should be acknowledged (see template below) and you must include, in appendix, the prompt, raw response, and your changelog how you have adopted the GenAI response in your work.* You must complete the work **independently**, and you must not share or show your work to anyone else, unless it is explicitly instructed in your task to do so. If you have questions or encounter problems, attend the on-campus or online classes, or use the **Discussion Board** on the unit site.

Per Deakin policy, include an Acknowledgements section:

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed June 2026): Used to *<generate structured critique on protocol design and confounding variables (3 iterations)>*.
- Microsoft Copilot for web (accessed June 2026): Used to *<summarise ASHRAE comfort standards and identify relevant literature.>*

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

Overview

Updating continuous data in dashboard for live monitoring is a challenging task. Two underlying tasks need to be handled elegantly, firstly, receiving data into a buffer and secondly, picking up data bits to update the dashboard graphics to have minimal changes so that user experiences a continuous data streaming like movies where the sequence of image frames create a sense of continuity.

Hardware Required

- i. A compatible smartphone – Android or iPhone.

Software Required

- i. Arduino IoT Cloud
- ii. Python 3,
- iii. Plotly Python library,
- iv. Numpy and Pandas Python library

Pre-requisites: You must do the following before this task

Week 8 activities in the unit site.

Task Objective

In Week 8 activity, you have connected your smartphone to Arduino IoT Cloud and streamed phone's accelerometer data to the cloud dashboard. You have also connected custom Python device where Python script can receive your phone's accelerometer data to visualise in Plotly Dash. The graphs in Dash were updated once enough data arrived to generate the next graph consisting of the same number of new N samples. Waiting for N number of new samples to create a new graph and updating the Dash would create discomfort to user due to observing the transition of graphs. Your job is to make the Dash graph update smoothly so that users do not notice transition of graph redrawing.

Steps:

1. Setup your smartphone, Arduino IoT cloud and Python script so phone's accelerometer data can flow through the Arduino IoT cloud to Python script.
2. Do necessary mechanism to smoothly project continuous accelerometer x, y, and z variables to Plotly Dash graphs.
3. The level of sophistication in handling continuous accelerometer data and updating the Dash graphs will reflect the scores to be awarded for this task. You are free to come up with alternative solutions to justify your technical competency. It may require hacking into Plotly library, should you require to update any of the library functions.
4. Wrap your work into a new function that can be used by your peers to achieve smooth Dash updates for any other continuous data without knowing much sophistication how continuous data is handled for live smooth monitoring. Justify the design and usage of your new function API.

Submission details

Q1. Describe your method to achieve smooth graphs update in Plotly Dash on arrival of fresh accelerometer data from your phone. Be descriptive enough and use visuals such as flow diagrams and tables, if required.

Q2. Describe your new wrapper function that you plan to release for Plotly Dash users to achieve smooth and real-time graphs update without knowing much about the complex data handling. This description should work as a documentation of your wrapper function.

Q3. Create a video in Panopto/CloudDeakin showing your program execution, continuous graph updates, your new function working and share the video link here.

Q4. Create a subdirectory 'week-8.2C' under directory 'SIT225_2024T2' in your drive where you copy the Python script file which contains Plotly Dash functions, Arduino sketch file if any, data file and the generated graphs. Commit and push to changes to GitHub. Include the link to your repository here with a GitHub page screenshot of weekly folder content. A tutor may try to access your GitHub link, if necessary. Give access to your tutor by adding tutor's email address as a collaborator of your **private** repository.

Instructions

Consolidate outputs following the submission details above into a single PDF file.

Submit your work

When you are ready, login to OnTrack and submit your pdf which consolidates all the items mentioned in the submission detail section above. Remember to save and backup your work.

Complete your work

After your submission, your OnTrack reviewer (tutor) will review your submission and give you feedback in about 5 business days. Your reviewer may further ask you some questions on the weekly topics and/or about your submissions. You are required to address your OnTrack reviewer's questions as a form of task discussions. Please frequently login to OnTrack for the task **Discuss/Demonstrate** or **Resubmit** equivalent to fix your work (if needed) based on the feedback to get your task signed as **Complete**.